Countable chain conditions for free quasi-Banach lattices: verification, a complete endpoint theorem, and further positive cases

Research note prepared from the supplied partial solution

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Status. This is a mathematical audit and research note, not a peer-reviewed paper. The general case for $0 < p < 1$ is not settled here. All claims labelled as new below should be independently checked before citation or submission.

Abstract

Let $0 < p \leq 1$, let E be a p-natural quasi-Banach space, put $E^{\triangle} = \mathcal{L}(E, L_p[0,1])$, and endow $B_{E^{\triangle}}$ with pointwise $L_p[0,1]$-norm convergence. Salguero-Alarcón, Tradacete, and Trejo-Arroyo asked whether the lattice $C^b_{\mathrm{ph}}(B_{E^{\triangle}}, L_p[0,1])$ always has the countable chain condition (CCC).

The supplied solution packet is substantially correct. Its product theorem for ℓ_p-sums of separable blocks is correct. Its theorem for bounded separable Boolean coordinate decompositions is also correct after one quantitative repair: the last error estimate must be made in the metric $d_p(u,v) = \|u - v\|_p^p$, rather than by adding ordinary $L_p[0,1]$-norm errors. This note gives a corrected proof.

A search completed on 22 June 2026 found no later proof or counterexample to the full conjecture. We then prove an abstract factorization–approximation criterion for the relevant "conic CCC." It yields a complete affirmative answer at the endpoint $p = 1$ (in fact for every quasi-Banach domain E), and, for $0 < p < 1$, an affirmative answer whenever E is a Banach space of nontrivial Rademacher type. The remaining nonlocally convex case is isolated, together with precise obstructions to the most direct extensions of the argument.

Contents

1 The problem and the outcome

Fix $0 < p \leq 1$ and write $Y = L_p[0,1]$. For a quasi-Banach space E let

$$E^{\triangle} = \mathcal{L}(E, Y), \qquad B_{E^{\triangle}} = \{T \in \mathcal{L}(E, Y) : \|T\| \leq 1\}.$$

The weak-$\triangle$ topology $\sigma(E^{\triangle}, E)$ on this ball is the initial topology for the evaluation maps

$$\mathrm{ev}_x : T \longmapsto Tx \in Y \qquad (x \in E),$$

where Y carries its quasi-norm topology. The source question is:

Question 1.1 (Salguero-Alarcón–Tradacete–Trejo-Arroyo)**.** *For every p-natural quasi-Banach space E, does*

$$C^b_{\mathrm{ph}}(B_{E^{\triangle}}, Y)$$

have lattice CCC?

Here lattice CCC means that every pairwise disjoint family of nonzero positive elements is countable. The source proves the assertion for $E = \ell_p(\Gamma)$ and observes that separable E are also covered.

The conclusions of this note are as follows.

(I) The supplied p-sum theorem is correct.

(II) The supplied Boolean-coordinate theorem is correct after a repair of its final p-metric estimate.

(III) No full answer or counterexample was located in the literature search completed on 22 June 2026.

(IV) Question 1.1 has an affirmative answer for $p = 1$, for every Banach space E; the proof actually works for every quasi-Banach domain.

(V) For $0 < p < 1$, Question 1.1 has an affirmative answer whenever E is a Banach space with nontrivial Rademacher type.

(VI) The general case $0 < p < 1$ remains open in this note.

The endpoint and nontrivial-type results come from Theorem 4.1, which may be useful beyond the present problem.

2 Cones, a countable directional cover, and Ramsey's lemma

It is helpful to separate the topological property actually used by homogeneous functions from ordinary topological CCC.

Definition 2.1. A subset $U \subseteq B_{E^\triangle}$ is a *relatively open cone* if it is relatively open and

$$T \in U, \quad \lambda > 0, \quad \lambda T \in B_{E^\triangle} \quad \Longrightarrow \quad \lambda T \in U.$$

We say that $B_{E^\triangle}$ has *conic CCC* if every pairwise disjoint family of nonempty relatively open cones is countable.

The following elementary cover replaces the ball construction in the source paper and is valid uniformly for all $0 < p \leq 1$.

Lemma 2.2 (Countable directional cone cover)**.** *There is a countable family $\mathcal{V} = (V_n)_{n\in\mathbb{N}}$ of relatively open cones in $Y_+ \setminus \{0\}$ such that*

$$Y_+ \setminus \{0\} = \bigcup_{n=1}^{\infty} V_n,$$

and whenever $u, v \in V_n$, one has $u \wedge v \neq 0$.

Proof. Let $S_+ = \{u \in Y_+ : \|u\|_p = 1\}$. It is separable, so choose a countable 1/3-net (b_n) in S_+. Put

$$V_n = \left\{ u \in Y_+ \setminus \{0\} : \left\| \frac{u}{\|u\|_p} - b_n \right\|_p < \frac{1}{3} \right\}.$$

These sets are open cones and cover $Y_+ \setminus \{0\}$. If $u, v \in V_n$ and $u \wedge v = 0$, their normalizations u_0, v_0 are disjoint positive unit vectors, hence

$$\|u_0 - v_0\|_p^p = \|u_0\|_p^p + \|v_0\|_p^p = 2.$$

On the other hand,

$$\|u_0 - v_0\|_p^p \leq \|u_0 - b_n\|_p^p + \|b_n - v_0\|_p^p < 2(1/3)^p < 2,$$

a contradiction. $\square$

Lemma 2.3 (Conic CCC lifts to the function lattice)**.** *If $B_{E^\triangle}$ has conic CCC, then $C^b_{\mathrm{ph}}(B_{E^\triangle}, Y)$ has lattice CCC.*

Proof. Suppose that $(f_i)_{i\in I}$ is an uncountable family of nonzero positive, pairwise disjoint functions. For each i choose T_i with $f_i(T_i) \neq 0$. By Lemma 2.2, after passing to an uncountable subfamily there is a fixed V_n such that $f_i(T_i) \in V_n$ for all i. Then

$$U_i = f_i^{-1}(V_n)$$

is a nonempty relatively open cone. The U_i are pairwise disjoint: a point in $U_i \cap U_j$ would give two values in the same V_n, hence with nonzero lattice infimum, contrary to $f_i \wedge f_j = 0$. This contradicts conic CCC. $\square$

For later use we record the finite combinatorial lemma used in the scalar-valued free Banach lattice argument of Avilés, Plebanek, and Rodríguez Abellán.

Lemma 2.4 (Finite directed Ramsey pattern)**.** *For every positive integer a there is $N = N(a)$ such that every map*

$$c : (N \times N) \setminus \Delta_N \longrightarrow \{1, \dots, a\}$$

admits $i < j < k$ with

$$c(i, j) = c(i, k), \qquad c(k, i) = c(k, j).$$

Proof. Apply the finite Ramsey theorem to triples, with a^2 colors, and obtain a five-element set on which

$$\{i < j < k\} \longmapsto (c(i, j), c(k, j))$$

is constant. If its elements are $b_1 < \cdots < b_5$, then $(i, j, k) = (b_2, b_3, b_4)$ has the required property: compare the triples $\{b_2, b_3, b_4\}$ and $\{b_2, b_4, b_5\}$ for the first equality, and $\{b_1, b_3, b_4\}$ and $\{b_1, b_2, b_4\}$ for the second. □

3 Audit of the source proof and of the supplied packet

3.1 Two repairable gaps in the printed source proof

The statement proved in the source paper for $E = \ell_p(\Gamma)$ is correct, but the printed proof contains two points that should not be reused verbatim when $p < 1$.

First, the proof of its Lemma 6.9 invokes the ordinary triangle inequality in L_p; for $p < 1$ only the p-triangle inequality is available. This is fixed by Lemma 2.2 (or by choosing radii depending on p and working with p-th powers).

Second, the proof of its Theorem 6.10 colors each pair of indices by one ball from a countable family and then extracts an uncountable subfamily with a common color. Such an extraction is not valid for arbitrary countable colorings of pairs. No pair-coloring is needed: for each nonzero function choose one nonzero value, use the countable cover to obtain a common cone on an uncountable subfamily, and then use the pairwise disjoint preimages exactly as in Lemma 2.3.

The source theorem therefore follows cleanly from its product identification

$$B_{\ell_p(\Gamma)^\triangle} \cong (B_{L_p[0,1]})^\Gamma,$$

the fact that an arbitrary product of separable metrizable spaces has (topological) CCC, and the preceding lifting argument.

3.2 The p-sum theorem in the packet

We now verify the first main mechanism in the supplied packet.

Lemma 3.1 (Operator ball of a p-sum)**.** *Let $(E_\gamma)_{\gamma\in\Gamma}$ be p-Banach spaces and*

$$E = \left(\bigoplus_{\gamma\in\Gamma} E_\gamma \right)_p .$$

Then restriction to the summands gives a homeomorphism

$$\Phi : B_{E^\triangle} \longrightarrow \prod_{\gamma\in\Gamma} B_{E_\gamma^\triangle}, \qquad \Phi(T) = (T \circ j_\gamma)_\gamma,$$

where all balls carry their pointwise topologies.

Proof. If $(T_\gamma)_\gamma$ belongs to the product and $x = (x_\gamma) \in E$, define

$$Tx = \sum_{\gamma \in \Gamma} T_\gamma x_\gamma.$$

The support of x is countable and finite partial sums are Cauchy because

$$\left\| \sum_{\gamma \in F} T_\gamma x_\gamma \right\|_p^p \leq \sum_{\gamma \in F} \|x_\gamma\|^p .$$

Thus T is a contraction and is the unique operator with the stated restrictions. This proves bijectivity.

Continuity of Φ is immediate. For continuity of the inverse, fix $x = (x_\gamma)$. Given $\varepsilon > 0$, choose finite $F \subseteq \Gamma$ with

$$2 \sum_{\gamma \notin F} \|x_\gamma\|^p < \varepsilon/2.$$

The finite-coordinate sum is continuous. If it changes by less than $\varepsilon/2$ in the metric $d_p(u, v) = \|u - v\|_p^p$, then the full sum changes by less than ε, using the p-triangle inequality and the two tail estimates. Hence every evaluation of the inverse is continuous. $\square$

Lemma 3.2 (Separable blocks)**.** *If E is separable, then $B_{E^\triangle}$ is separable and metrizable in the pointwise topology.*

Proof. Let (x_n) be dense in E. On the operator unit ball, convergence at every x_n implies convergence at every $x \in E$, uniformly through

$$\|T(x - x_n)\|_p \leq \|x - x_n\| .$$

Thus the ball embeds topologically in the countable product $Y^{\mathbb{N}}$. The latter is separable metrizable and second countable; every subspace is therefore separable metrizable. $\square$

Theorem 3.3 (Verified p-sum case)**.** *Let $(E_\gamma)_{\gamma \in \Gamma}$ be separable p-natural quasi-Banach spaces and put*

$$E = \left(\bigoplus_{\gamma \in \Gamma} E_\gamma \right)_p .$$

Then E is p-natural and $C^b_{\mathrm{ph}}(B_{E^\triangle}, L_p[0, 1])$ has CCC.

Proof. The coordinatewise sum of witnessing p-convex p-Banach lattices witnesses that E is p-natural. By Lemmas 3.1 and 3.2, $B_{E^\triangle}$ is homeomorphic to a product of separable metrizable spaces. Such a product has topological CCC (use the Δ-system lemma on the finite supports of basic open sets and countability of the bases on the root). Topological CCC implies conic CCC, and Lemma 2.3 finishes the proof. $\square$

This simultaneously recovers the source case $E = \ell_p(\Gamma)$ and the separable case.

3.3 Corrected Boolean-coordinate theorem

We use the packet's terminology. A p-Banach space E has a *bounded separable Boolean coordinate decomposition* over Γ if there are projections $(P_A)_{A\subseteq\Gamma}$ with

$$P_\varnothing = 0, \quad P_\Gamma = I, \quad P_A P_B = P_{A\cap B},$$

$$P_{A\cup B} = P_A + P_B \quad (A\cap B=\varnothing),$$

finite-coordinate convergence $P_F x \to x$, separable one-coordinate ranges, and

$$K = \sup_{A\subseteq\Gamma} \|P_A\| < \infty.$$

Theorem 3.4 (Corrected coordinate-decomposition case)**.** *Let $0<p\le 1$. If a p-natural quasi-Banach space E has a bounded separable Boolean coordinate decomposition, then $C^b_{\mathrm{ph}}(B_{E^\triangle}, L_p[0,1])$ has CCC.*

Proof. It suffices, by Lemma 2.3, to prove conic CCC. Suppose that $(U_i)_{i\in I}$ is an uncountable pairwise disjoint family of nonempty relatively open cones in $B_{E^\triangle}$. Choose $\eta>0$ so small that

$$(2^p+2)(K\eta)^p < 1. \tag{3.1}$$

Finite-block centers. For $T\in B_{E^\triangle}$, the operators $K^{-1}TP_F$ belong to the unit ball and converge pointwise to $K^{-1}T$. Since each U_i is an open cone, we can therefore choose

$$S_i\in U_i, \qquad \|S_i\|<\eta, \qquad S_i = S_i P_{A_i}$$

for some finite $A_i\subseteq\Gamma$. Choose a basic neighborhood

$$N_i = \{T\in B_{E^\triangle} : d_p(Tx_{i,k}, S_i x_{i,k}) < \varepsilon_i,\ 1\le k\le m_i\} \subseteq U_i, \tag{3.2}$$

where $d_p(u,v) = \|u-v\|_p^p$. Enlarge A_i, without changing $S_i = S_i P_{A_i}$, so that

$$\left\|P_{\Gamma\setminus A_i} x_{i,k}\right\|^p < \frac{\varepsilon_i}{4(\eta K)^p} \quad (1\le k\le m_i). \tag{3.3}$$

Δ-*system and root operators.* After passing to an uncountable subfamily, the finite sets A_i form a Δ-system with root R: the tails $A_i\setminus R$ are pairwise disjoint. The range $E_R = P_R E$ is separable, being a finite sum of separable coordinate ranges. The set

$$\mathcal{R} = \{R_0\in\mathcal{L}(E_R, L_p[0,1]) : \|R_0\| < 2\eta\},$$

with pointwise convergence on E_R, is separable metrizable. For each i let $O_i\subseteq\mathcal{R}$ consist of those R_0 satisfying

$$d_p\big((R_0P_R + S_iP_{A_i\setminus R})x_{i,k}, S_i x_{i,k}\big) < \varepsilon_i/4 \quad (1\le k\le m_i). \tag{3.4}$$

Each O_i is nonempty and relatively open: it contains the restriction of S_i to E_R. Hence two of them, say O_i and O_j, intersect. Fix $R_0\in O_i\cap O_j$.

Gluing. Define

$$S = R_0P_R + S_iP_{A_i\setminus R} + S_jP_{A_j\setminus R}.$$

The three coordinate pieces are disjoint, so for $\|x\|\le 1$,

$$\|Sx\|_p^p \le (2\eta K)^p + 2(\eta K)^p = (2^p+2)(K\eta)^p < 1$$

by (3.1). Thus $S \in B_{E^\triangle}$.

For the i-th neighborhood, the root and i-tail part has d_p-error less than $\varepsilon_i/4$ by (3.4). The only additional term is the j-tail. Since $A_j \setminus R \subseteq \Gamma \setminus A_i$,

$$\left\| S_j P_{A_j \setminus R} x_{i,k} \right\|_p \leq \eta K \left\| P_{\Gamma \setminus A_i} x_{i,k} \right\|,$$

so its d_p-error is less than $\varepsilon_i/4$ by (3.3). The p-triangle inequality therefore gives total d_p-error less than $\varepsilon_i/2 < \varepsilon_i$. Thus $S \in N_i$. The same argument gives $S \in N_j$, contradicting disjointness of U_i and U_j. $\square$

The quantitative issue in the supplied proof occurs precisely in its last line: errors bounded by $\varepsilon_i/2$ and $\varepsilon_i/8$ cannot be added as ordinary norms when $p < 1$. Equations (3.3)–(3.4) repair this by reserving p-metric error $\varepsilon_i/4$ for each contribution.

Corollary 3.5. *Let $(E_\gamma)_{\gamma\in\Gamma}$ be separable p-natural spaces. If $p \leq r < \infty$, then Question 1.1 has an affirmative answer for*

$$\left(\bigoplus_{\gamma\in\Gamma} E_\gamma\right)_r \quad \text{and} \quad \left(\bigoplus_{\gamma\in\Gamma} E_\gamma\right)_{c_0}.$$

In particular it holds for $\ell_r(\Gamma)$, $p \leq r < \infty$, and for $c_0(\Gamma)$.

Proof. The usual coordinate projections are contractive and have the required finite convergence. The sums are p-natural. For the finite r-sum this follows from Minkowski's inequality in $\ell_{r/p}$; for the c_0-sum it follows by taking coordinatewise suprema. Apply Theorem 3.4. $\square$

4 A factorization–approximation criterion

The next result is the main new mechanism in this note. It turns a uniform factorization through a space with finite-rank approximations into conic CCC. The proof combines strong approximation, norm separability of the left factors, and Lemma 2.4.

Theorem 4.1 (Factorization–approximation criterion)**.** *Let $0 < p \leq 1$, $Y = L_p[0,1]$, and let E be a quasi-Banach space. Suppose there exist a quasi-Banach space Z, finite-rank operators $Q_n : Z \to Z$, a set $\mathcal{A} \subseteq \mathcal{L}(Z, Y)$, and constants $\Lambda, M, C < \infty$ such that:*
(a) *$\sup_n \|Q_n\| \leq \Lambda$ and $Q_n z \to z$ for every $z \in Z$;*
(b) *$\mathcal{A}$ is separable in operator norm and $\sup_{A\in\mathcal{A}} \|A\| \leq M$;*
(c) *every $T \in \mathcal{L}(E, Y)$ factors as $T = AS$, where $A \in \mathcal{A}$, $S \in \mathcal{L}(E, Z)$, and $\|S\| \leq C\|T\|$.*
Then $B_{E^\triangle}$ has conic CCC. Consequently, $C^b_{\mathrm{ph}}(B_{E^\triangle}, Y)$ has lattice CCC.

Proof. Assume that $(U_i)_{i\in I}$ is an uncountable pairwise disjoint family of nonempty relatively open cones. Discarding at most one member, none contains zero. Choose $\rho > 0$ with

$$3(M\Lambda C\rho)^p < 1. \tag{4.1}$$

By radial invariance, choose $T_i \in U_i$ with $0 < \|T_i\| < \rho$. There is a basic neighborhood

$$N_i = \{T \in B_{E^\triangle} : d_p(Tx_{i,k}, T_i x_{i,k}) < \varepsilon_i,\ 1 \leq k \leq m_i\} \subseteq U_i. \tag{4.2}$$

After normalizing the test vectors, thinning the family, and decreasing the radii if necessary, we may suppose

$$m_i = m,\ \|x_{i,k}\| \leq 1, \quad \text{and} \quad \varepsilon_i > \varepsilon > 0 \tag{4.3}$$

for all i, k.

Factor $T_i = A_i S_i$ as in (c). Hence

$$\|S_i\| \le C\rho. \tag{4.4}$$

Set $\delta = \varepsilon/4$. By (a), for each i there is n_i such that

$$d_p(A_i Q_{n_i} S_i x_{i,k}, T_i x_{i,k}) < \delta \quad (1 \le k \le m).$$

Passing to an uncountable subfamily, take a common n and put

$$\widetilde{T}_i = A_i Q_n S_i.$$

Then

$$\left\|\widetilde{T}_i\right\| \le M\Lambda C\rho. \tag{4.5}$$

We next obtain a finite partition for all cross-evaluations. Put $F = Q_n Z$ and $R = \Lambda C\rho$. Since $\mathcal{A}$ is norm separable, choose $\theta > 0$ and pass to an uncountable subfamily on which

$$\|A_i - A_0\| < \theta \tag{4.6}$$

for a fixed $A_0 \in \mathcal{A}$. Choose θ so small that $\theta^p R^p < \delta/4$. The set

$$K_0 = A_0\{z \in F : \|z\| \le R\}$$

is compact. Choose a finite d_p-net in K_0 of radius $\delta/4$. For $a \ne b$ and $1 \le k \le m$, write

$$y_{a,b,k} = \widetilde{T}_b x_{a,k} = A_b Q_n S_b x_{a,k}.$$

The vector $z = Q_n S_b x_{a,k}$ lies in F and has norm at most R. By (4.6), $d_p(y_{a,b,k}, A_0 z) < \delta/4$. Assign $y_{a,b,k}$ to one of the finitely many net points within d_p-distance $< \delta/2$. Two values assigned to the same point have mutual d_p-distance $< \delta$.

Thus every ordered pair (a, b) receives a color $c(a, b)$, namely the m-tuple of cells containing $(y_{a,b,k})_{k=1}^m$, from a fixed finite set. Choose finitely many indices of cardinality N from Lemma 2.4. That lemma gives $i < j < k$ with

$$c(i, j) = c(i, k), \qquad c(k, i) = c(k, j). \tag{4.7}$$

Define

$$R_{ijk} = \widetilde{T}_i - \widetilde{T}_j + \widetilde{T}_k.$$

By (4.1) and (4.5),

$$\|R_{ijk}\|^p \le \sum_{\ell \in \{i,j,k\}} \left\|\widetilde{T}_\ell\right\|^p < 1,$$

so $R_{ijk} \in B_{E^\triangle}$.

For an i-test vector, (4.7) and the construction of the cells give

$$d_p(\widetilde{T}_k x_{i,r}, \widetilde{T}_j x_{i,r}) < \delta.$$

Together with the own-center approximation, this yields

$$\begin{aligned} d_p(R_{ijk} x_{i,r}, T_i x_{i,r}) &\le d_p(\widetilde{T}_i x_{i,r}, T_i x_{i,r}) + d_p(\widetilde{T}_k x_{i,r}, \widetilde{T}_j x_{i,r}) \\ &< 2\delta = \varepsilon/2 < \varepsilon_i. \end{aligned}$$

Hence $R_{ijk} \in N_i$. The second equality in (4.7) gives, identically, $R_{ijk} \in N_k$. This contradicts $N_i \subseteq U_i$, $N_k \subseteq U_k$, and disjointness. Conic CCC follows, and Lemma 2.3 gives lattice CCC. $\square$

5 Consequences: the endpoint and nontrivial type

5.1 A complete affirmative answer for $p = 1$

Theorem 5.1 (Complete endpoint theorem)**.** *For every quasi-Banach space E,*

$$C^b_{\mathrm{ph}}\big(B_{\mathcal{L}(E,L_1[0,1])}, L_1[0,1]\big)$$

has lattice CCC. In particular, Question 1.1 has an affirmative answer for $p = 1$ and every Banach space E.

Proof. Apply Theorem 4.1 with

$$Z = L_1[0,1], \qquad \mathcal{A} = \{\mathrm{Id}_{L_1[0,1]}\}, \qquad A = \mathrm{Id}, \qquad S = T.$$

Let Q_n be conditional expectation onto the dyadic partition of level n. Then Q_n has finite-dimensional range, $\|Q_n\| = 1$, and $Q_n f \to f$ in L_1 for every $f \in L_1$. Thus all hypotheses hold with $\Lambda = M = C = 1$. $\square$

The proof uses no local convexity of the domain. The endpoint is special because L_1 has contractive finite-rank approximations. This feature disappears for the atomless $L_p[0,1]$ when $0 < p < 1$; see Section 6.

5.2 Banach spaces of nontrivial type

We use Maurey's factorization theorem in the following standard form. If a Banach space E has Rademacher type $q > r$, and $0 < p < r$, then every $T : E \to L_p(\mu)$ factors

$$T = M_g S,$$

where $S : E \to L_r(\mu)$ is bounded, $g \geq 0$ belongs to $L_s(\mu)$, $1/p = 1/r + 1/s$, $\|g\|_s = 1$, and $\|S\| \leq C\,\|T\|$ with C depending only on the parameters and E. This is the form of Maurey's theorem used in Section 5 of the source paper.

Theorem 5.2 (Nontrivial-type Banach domains)**.** *Let $0 < p < 1$. If E is a Banach space with Rademacher type $q > 1$, then*

$$C^b_{\mathrm{ph}}\big(B_{\mathcal{L}(E,L_p[0,1])}, L_p[0,1]\big)$$

has lattice CCC.

Proof. Apply Maurey's theorem with $r = 1$. Put $s = p/(1-p)$, so $1/p = 1 + 1/s$. Every $T : E \to L_p[0,1]$ factors as

$$T = M_g S, \qquad S : E \to L_1[0,1], \qquad \|S\| \leq C\,\|T\|,$$

where $g \geq 0$ and $\|g\|_s = 1$. Let

$$\mathcal{A} = \{M_g : L_1[0,1] \to L_p[0,1] : g \in L_s,\ g \geq 0,\ \|g\|_s = 1\}.$$

Generalized Hölder gives

$$\|M_g f\|_p \leq \|g\|_s\,\|f\|_1, \qquad \|M_g - M_h\|_{\mathcal{L}(L_1,L_p)} \leq \|g - h\|_s.$$

Since $L_s[0,1]$ is separable for every finite $s > 0$, $\mathcal{A}$ is bounded and operator-norm separable. Take $Z = L_1$ and the dyadic conditional expectations as in Theorem 5.1. Theorem 4.1 applies. $\square$

Every Banach space is p-natural for $0 < p \leq 1$ (embed it isometrically in a $C(K)$-space), so Theorem 5.2 gives genuine cases of the source conjecture. It includes, for example, Hilbert spaces and L_r-spaces for $1 < r < \infty$, without separability assumptions.

6 Further reductions and genuine obstructions

6.1 The algebraic operator space is dense and harmless

Fix a Hamel basis H of a p-natural space E. Algebraic linear maps $E \to L_p[0,1]$ identify with the product $(L_p[0,1])^H$. The source paper's finite interpolation theorem (its Theorem 4.6) says that, for every finite linearly independent set in E, arbitrary $L_p[0,1]$-values are realized by a bounded operator. Hence

$$E^{\triangle} = \mathcal{L}(E, L_p[0,1])$$

is dense, for pointwise convergence, in the full algebraic operator space. The product $(L_p[0,1])^H$ has topological CCC. This does not settle the problem: the unit ball is a subspace cut out by a global norm constraint, and topological CCC is not hereditary. The unresolved difficulty is therefore quantitative, not finite-algebraic.

6.2 Why separability of L_p alone is insufficient

The finite-color hypothesis in Lemma 2.4 cannot be replaced by "countably many colors." Indeed, for each countable ordinal $\beta < \omega_1$ choose an injection $e_\beta : \beta \to \mathbb{N}$ and define, for $\alpha < \beta$,

$$c(\beta, \alpha) = e_\beta(\alpha).$$

Define the remaining directed colors arbitrarily. Then for every $i < j < k$,

$$c(k, i) \neq c(k, j),$$

so the Ramsey pattern fails.

This obstruction can be encoded by actual operators. Choose a separated sequence (y_n) in $B_{L_p[0,1]}$ and, on $E = \ell_p(\omega_1)$, prescribe

$$T_\beta e_\alpha = y_{c(\alpha,\beta)}.$$

The p-sum estimate makes each T_β a contraction. Thus an argument that only replaces the bounded L_p-ball by a countable metric coloring cannot prove the conjecture, even in a case where the conjecture is known to hold. The finite-rank step in Theorem 4.1 is exactly what turns the relevant cross-values into finitely many small-diameter cells.

6.3 Why the $p = 1$ proof does not directly extend

For atomless $L_p[0,1]$ with $0 < p < 1$, the continuous dual is trivial. Therefore every finite-rank continuous endomorphism of $L_p[0,1]$ is zero: the coordinate functionals of a finite-rank map would be continuous linear functionals on L_p. In particular, there cannot be a nontrivial finite-rank approximation of the identity analogous to the dyadic conditional expectations on L_1. This is the structural reason Theorem 5.1 stops at the endpoint.

Theorem 4.1 bypasses this only when operators factor through a different space, such as L_1, that does have finite-rank approximations. For general p-natural quasi-Banach domains no such uniform factorization is known.

6.4 What a remaining counterexample would have to avoid

Combining the source paper, the verified packet, and the new results above, any counterexample known to this note must satisfy all of the following:

- $0 < p < 1$;
- E is nonseparable;
- E is not covered by a bounded separable Boolean coordinate decomposition;
- the relevant operators do not admit the uniform factorization scheme of Theorem 4.1; in particular, in the Banach case the space cannot be handled by the nontrivial-type factorization used in Theorem 5.2.

No construction meeting these requirements and producing uncountably many disjoint open cones was found.

7 Literature status as of 22 June 2026

The exact-title, arXiv-identifier, author-page, citation, and phrase searches performed for this audit found the following.

- The source remains arXiv:2512.05273v1, posted 4 December 2025.
- Current author publication pages still list it as a preprint.
- A February 2026 workshop on free Banach lattices included talks on free quasi-Banach lattices and open-problem sessions, but no posted solution to Question 1.1 was located.
- A 2026 paper located by citation search merely cites the preprint and does not address this question.
- The closest earlier result is the scalar-valued theorem of Avilés, Plebanek, and Rodríguez Abellán: $C_{\mathrm{ph}}(B_{E^*})$ satisfies the stronger σ-bounded chain condition for every Banach space E. Its finite directed Ramsey lemma is the combinatorial input in Theorem 4.1.

Accordingly, no full published or posted proof or counterexample was found. This is a report of the search, not a claim about unpublished work or material outside the searched sources.

8 Final verdict

Claim or argument	Verdict	Required action / comment
Countable cone cover and lifting	Correct	Use p-th powers; avoids the source ball-radius issue.
p-sum/product homeomorphism	Correct	Infinite-support continuity estimate is valid.
Separable block ball is separable metrizable	Correct	It is a subspace of a countable product of separable metric spaces.
p-sums of separable blocks	Correct	Gives topological CCC and hence lattice CCC.
Boolean-coordinate theorem	Correct after repair	Replace the final ordinary-norm error addition by the d_p estimates in (3.3)–(3.4).
ℓ_r- and c_0-sum corollaries	Correct	Coordinate projections are contractive; p-naturality is preserved.
General conjecture for $p = 1$	Affirmative here	Theorem 5.1.
General conjecture for $0 < p < 1$	Not settled	No counterexample found; Theorems 3.4 and 5.2 give substantial positive classes.

The supplied partial result is therefore worth retaining, but its coordinate proof should be replaced by the corrected version above. The most substantial additional conclusion is the complete endpoint theorem $p = 1$. For $0 < p < 1$, the factorization criterion isolates a concrete route to further progress: find a uniform factorization of $\mathcal{L}(E, L_p)$ through a space with finite-rank approximations and with an operator-norm separable family of left factors.

References

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